How to cite: *Angew. Chem. Int. Ed.* **2025**, e21038
doi.org/10.1002/anie.202521038

Hybrid Supercapacitors

Hydrophobic Association Enabled Salt-Concentrated all-Cellulose Hydrogel Electrolytes for High-Temperature Zinc-Ion Hybrid Supercapacitors

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Abstract: Hydrogel electrolytes have great potential in electrochemical energy storage devices due to their high flexibility, conductivity, and safety. However, at high temperatures, the mechanical stability and ionic conductivity of hydrogel electrolytes are compromised due to water evaporation and unstabilization of polymer matrix. Herein, we propose a salt-concentrated hydrogel electrolyte with its matrix showing hydrophobic association effect upon heating for achieving high-temperature adaptability. The cotton cellulose supports a stable homogeneous polymer-salt phase in the hydrogel electrolytes. The methylcellulose works through temperature-dependent hydrophobic interactions, maintaining the stability of the salt-concentrated hydrogel over a wide temperature range. An all-biobased hydrogel with both mechanical and conductive stability is realized at high temperatures (more than 60 °C). We have demonstrated a zinc-ion hybrid supercapacitor with a high specific capacity of 250.3 mAh g^{-1} at 120 °C and 85.4% retention for 10,000 cycles at 60 °C. This work is expected to provide a general strategy to achieve high-temperature hydrogel electrolyte.

Introduction

The emergence of flexible electronics technology has driven applications such as wearable electronics and portable medical devices.[1] Flexible energy storage devices are crucial for flexible electronics.[2] In addition to being lightweight, bendable and stretchable, the effective operation of flexible energy storage devices across diverse temperature conditions is a considerable challenge.[3] Among various electrochemical energy storage systems, aqueous zinc-ion hybrid supercapacitors (ZHSs) based on hydrogel electrolytes have garnered considerable interest because of their inherent safety, high energy and power densities, and long cycle life.[4–7] In these devices, the hydrogel electrolyte plays a dual role by providing efficient ionic conduction and absolute electronic insulation between the anode and cathode electrodes, which is critical for stable energy conversion.[8–11]

For specific application environments such as fire-fighting and space exploration, the stability of the flexible energy storage devices within a wide temperature window is essential.[12,13] In conventional hydrogel electrolytes, high temperatures accelerate water evaporation and lead to an irreversible loss of flexibility and ionic conductivity.[14] Currently, strategies used to inhibit water evaporation mainly involve increasing the number of hydrophilic groups with hydrophilic polymers or organic solvents.[15–19] Unfortunately, the ratio of organic substance in hydrogel is usually limited due to the need to maintain ionic conductivity. These hydrogel electrolytes still lack long-term water retention ability at high temperatures with unstable hydrophilic interactions.[20] The hydrogel electrolytes used in energy storage devices, including zinc-based capacitors and batteries have struggled to maintain long-term cycling stability beyond 60 °C.[15–17,21–23] The organic solvents in hydrogel electrolyte also increase the risk of toxicity and flammability. Introducing highly concentrated salt to hydrogel is an efficient approach to stabilizing water molecules and simultaneously inhibit side reactions, thus enabling high-temperature electrolyte with wider electrochemical window.[24,25] Currently, the fabrication of salt-concentrated hydrogel electrolytes that can resist high temperatures (more than 60 °C) has been less successful with the challenges in maintaining uniformity and mechanical integrity.[26,27] Fundamentally, the dispersibility of the polymer is severely restricted by competitive hydration of highly concentrated salt ions.[28] Furthermore, most of the non-covalent cross-linking mechanisms that are indispensable to the mechanical stability of hydrogels (e.g., hydrogen bonding, electrostatic interaction, and metal-ligand coordination) are weak or ineffective under the coupled high-temperature and salt-concentrated environments.[29–33]

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Hydrophobic association, which describes the tendency of nonpolar molecules or groups to cluster together in water, has been widely found to generate favorable mechanical properties in natural organisms.[34] For example, collagen proteins in animal skin and connective tissues require correct folding driven by hydrophobic associations to provide mechanical strength and toughness.[35] Noteworthy, the hydrophobic association provides an intrinsic potential of dynamic strengthening in response to increasing temperature.[36,37] Theoretically, combining hydrophobic association with highly concentrated salt ions presents a promising strategy for the hydrogel electrolytes to achieve thermal stability. However, to our knowledge, the fabrication of uniform salt-concentrated hydrogels with hydrophobic effect has been barely achieved. This is potentially due to the challenges in selecting proper materials and developing protocols to prevent the salting-out effect in such systems.

Herein, we report an all-cellulose-based hydrogel with concentrated zinc chloride ($ZnCl_2$) which shows high thermal stability. The highly concentrated $ZnCl_2$ supported the high reversibility of zinc deposition and stripping.[38,39] Bio-based renewable cellulose with abundant hydroxyl groups is able to participate in Zn^{2+} solvation and allows for a uniform polymer-salt phase.[40–42] By adding an amphiphilic methylcellulose, the temperature-dependent hydrophobic association is endowed,[43,44] thus enabling the mechanical and electrochemical properties to be stable over a wide temperature window. As a proof-of-concept, we assembled the $ZnCl_2$-concentrated hydrogel electrolyte into a ZHS. It shows a wide voltage window of 2.0 V. Noteworthy, the ZHS could reach a specific capacity of 250.3 mAh g^{-1} at 120 °C and exhibit a stable 10,000 cycle life at 60 °C with an 85.4% capacity retention. As the hydrophobic association effect and highly concentrated salt ions do not only occur in specific material systems, this work is expected to provide general guidance on achieving thermally stable hydrogels for various applications.

Results and Discussion

The failure (losing ionic conductivity and mechanical integrity) of conventional hydrogel electrolytes at high temperatures is mainly due to water evaporation and collapse of the matrix structure.[14] Accordingly, we have designed the targeted hydrogel electrolytes on the basis of the properties of selective salts and matrix polymers. We first demonstrated that the highly water-soluble $ZnCl_2$ can effectively inhibit water evaporation at its high concentrations. The water retention ability of $ZnCl_2$ aqueous solutions with different concentrations was tested at an elevated temperature of 80 °C (Figure S1). At low $ZnCl_2$ concentrations (≤35 wt%), the solution experienced rapid water evaporation. As the $ZnCl_2$ concentration increased, the water retention ratio of the solution improved significantly. Specifically, a 65 wt% $ZnCl_2$ solution retained 88% of its initial water content after standing at 80 °C for 14 h, while a 73 wt% $ZnCl_2$ solution showed no observable loss of water under identical conditions. This exceptional water retention of solution with highly concentrated $ZnCl_2$ highlights its potential as a stable electrolyte under high temperature environments. Noteworthily, methylcellulose was selected as a crucial matrix polymer to reinforce the hydrogel. Methylcellulose features abundant hydrophobic methyl groups ($-CH_3$) along its polymer chain (Fourier transform infrared spectroscopy, FTIR, Figure S2).[45] The methylcellulose chain would undergo supramolecular aggregation in aqueous environments at elevated temperatures due to the enhanced hydrophobic interactions of methyl groups, thus enabling temperature-dependent strengthening of the matrix (Figure S3a).[36] Digital pictures of the aqueous methylcellulose solutions show reversible transition between gel state at high temperature and liquid state at room temperature (Figure S3b). The unique thermally induced hydrophobic association of methylcellulose is expected to maintain the mechanical integrity of the hydrogel electrolyte under elevated temperature conditions.

Methylcellulose is easily soluble in water due to its amphiphilic nature and the formation of a water cage structure surrounding the methyl groups.[43] It is also dispersible in $ZnCl_2$ aqueous solution to form a homogeneous phase when the ion concentration is below 50 wt% (Figure S4). To stabilize the methylcellulose in highly concentrated hydrogel, a cotton linter cellulose with abundant hydroxyl groups was used together to construct the polymer matrix. Cotton linter cellulose has longer fibers than methylcellulose (Scanning electron microscopy, SEM, Figure S5). In contrast to methylcellulose, cotton linter cellulose has a highly crystalline structure (X-ray diffraction, XRD, Figure S6) and is therefore insoluble in water. However, it dissolves in highly concentrated zinc salts (65–73 wt%), which break the intermolecular hydrogen bonds and subsequently coordinate with the hydroxyl groups.[46,47] Meanwhile, the highly concentrated zinc ions are trapped within the cellulose chain and form a homogeneous phase.[28] With the existence of cotton linter cellulose, the methylcellulose can be also dispersed in the $ZnCl_2$ solution with a high concentration of 73 wt% (Figure S7). The large number of hydroxyl groups in long cellulose fibers contributes hydrogen bonds and coordination bonds to stabilize both the methylcellulose and the ions.

In our design, methylcellulose and cotton cellulose were combined to create a hydrogel matrix trapped with highly concentrated $ZnCl_2$, forming a stable hydrogel electrolyte over a wide temperature range. The methylcellulose/cellulose in hydrogel can still be identified by FTIR spectrum (Figure S8). The viscosities of hydrogel precursor solutions (cellulose/methylcellulose/$ZnCl_2$, cellulose/$ZnCl_2$ and $ZnCl_2$ alone) were measured as a function of temperature. Above 40 °C, the cellulose/methylcellulose/$ZnCl_2$ system exhibites less viscosity loss than cellulose/$ZnCl_2$, which can be attributed to the hydrophobic association effect of methylcellulose (Figure S9a). Notably, its viscosity remains nearly constant within a certain elevated temperature range, suggesting an enhanced hydrophobic interaction with temperature. Moreover, the cellulose/methylcellulose/$ZnCl_2$ solution displays excellent viscosity recovery over a thermal cycling between 25 and 100 °C, indicating the potential reversibility of the hydrogel matrix (Figure S9b). As explained in Figure 1a, the methylcellulose containing partial methyl groups on its chain is not close to the zinc ions with coordinated water,

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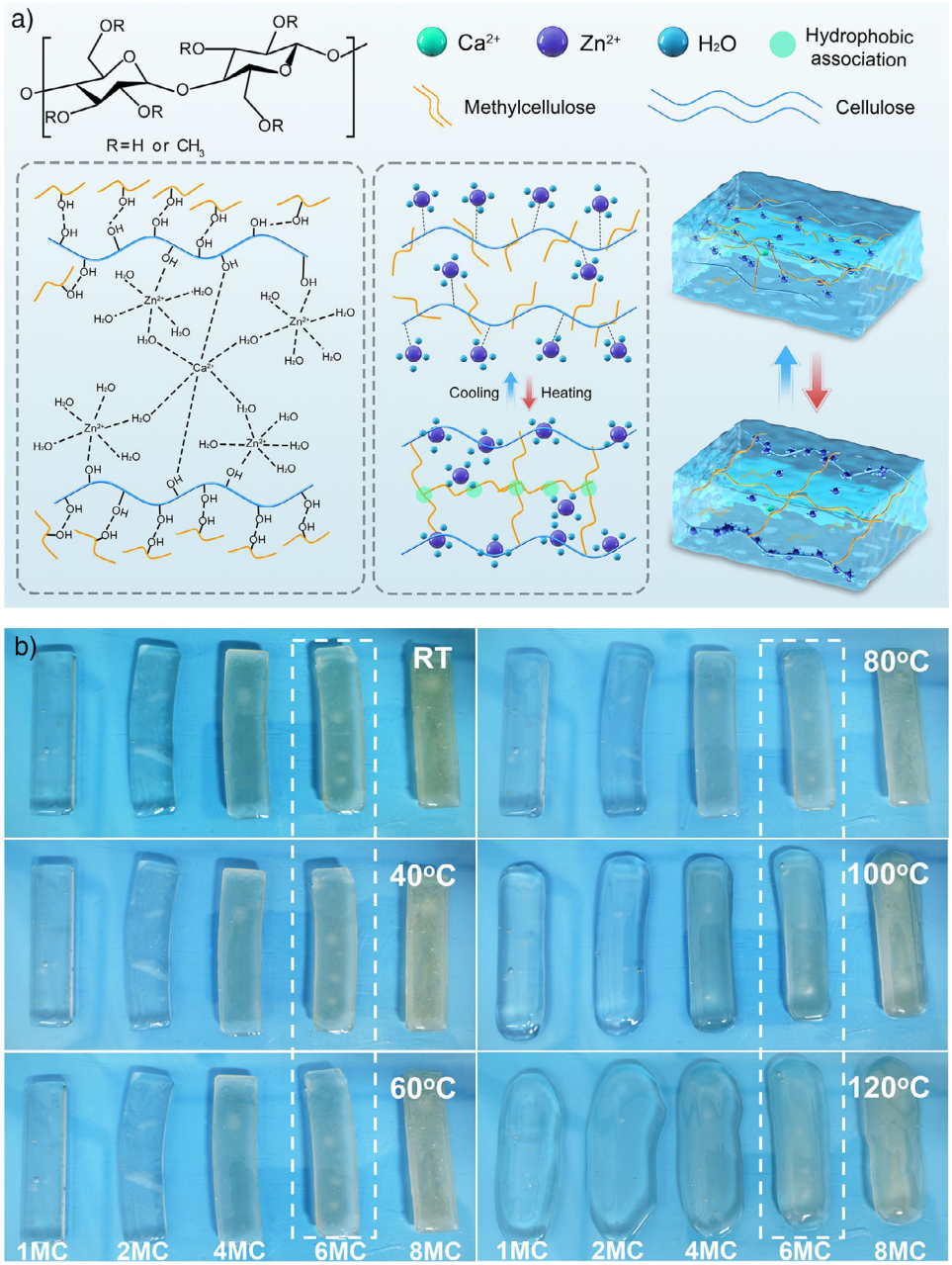

Figure 1. a) Schematic illustration of the design strategy of high-temperature-capable cellulose hydrogel electrolyte; b) photos of cellulose hydrogel electrolytes with different methylcellulose contents (CL:xMC) from room temperature (RT) to 120 °C high temperature.

while they are both stabilized by the abundant hydroxyl groups on cellulose. The cotton cellulose and methylcellulose are cross-linked through hydrogen bonds and further cross-linked by the coordination of Zn^{2+} and Ca^{2+} (The actual molar ratio of Zn/Ca is 30:1). As the temperature increases, the hydrogen bonding and metal-ligand coordination cross-linking network is broken with the disturbance of zinc ions.[40] At the same time, the hydrogel matrix is also reconfigured through dynamically strengthened hydrophobic interactions between the methylcelluloses. In this equilibrium, the hydrogel matrix remains stable. As shown in Figure 1b, the hydrogel electrolytes (CL:xMC) with cellulose and methylcellulose as the matrix remains largely intact at temperatures up to 80 °C. Such improved thermal mechanical stability when introducing more methylcellulose into the hydrogel can be directly visualized. Specifically, no observable collapse occurs for CL:6MC even at temperatures exceeding 100 °C. For comparison, a typical hydrogel electrolyte (CL:0MC) based on pure cellulose shows observable melting at 60 °C (Figure S10). The CL:xMC can maintain uniformity from room temperature to 120 °C high temperature without macroscopic phase separation, benefitting from the instantaneous weaving of supramolecular networks by hydrophobically associating polymers.[48] The hydrogel electrolyte (CL:6MC) experiences negligible water loss below 60 °C due to its highly concentrated salt ions (Figure S11).

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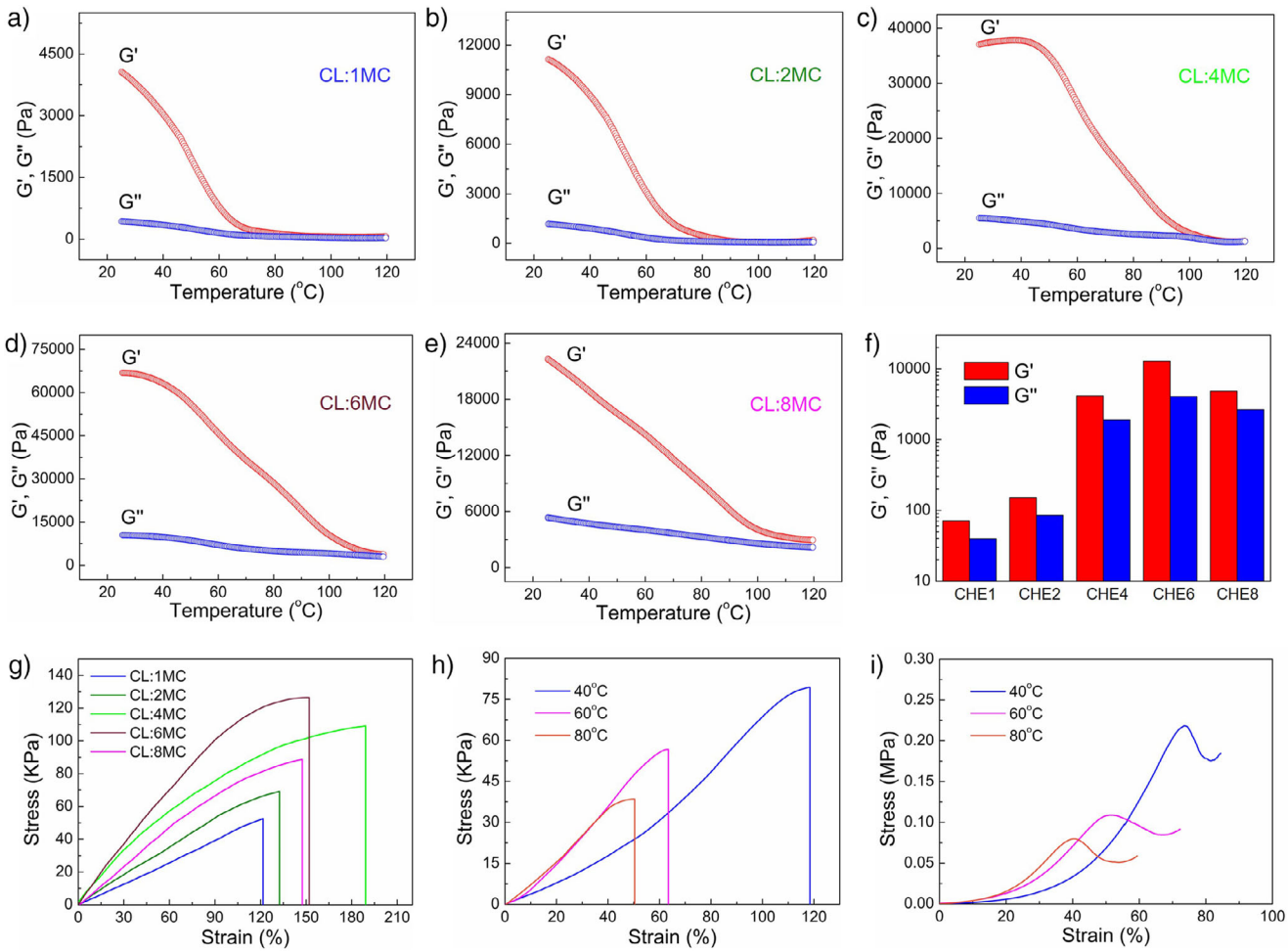

Figure 2. Temperature scanning rheological curves of a) CL:1MC, b) CL:2MC, c) CL:4MC, d) CL:6MC, and e) CL:8MC hydrogels; f) average values of G′ and G″ of CL:xMC in the temperature range of 80–120 °C from rheological tests; g) tensile stress-strain curves of CL:xMC hydrogels at room temperature; h) tensile stress-strain curves of CL:6MC hydrogel at 40–80 °C; i) compressive stress-strain curves of CL:6MC hydrogel at 40–80 °C.

Rheological tests were performed to further explore the thermal stability of the cellulose hydrogel electrolytes (CL:xMC) from room temperature to 120 °C (Figure 2a–e). Below 80 °C, the hydrogels exhibit significantly greater storage modulus G′ than loss modulus G″, indicating solid-state properties. Above 80 °C, the G′ is still greater than G″, especially for the hydrogels of CL:4MC, CL:6MC and CL:8MC. The bar chart provides a direct comparison for the average rheological G′ and G″ values of the hydrogels in the temperature range of 80–120 °C (Figure 2f). As the ratio of methylcellulose in hydrogel increases, both G′ and G″ increase from CL:1MC to CL:6MC and decrease for CL:8MC. Furthermore, the ratio of average rheological G′ values over the average rheological G″ values for CL:1MC, CL:2MC, CL:4MC, CL:6MC, and CL:8MC are 1.79, 1.78, 2.20, 3.15, and 1.83, respectively. CL:6MC has the highest rheological G′ value, which is significantly greater than its rheological G″ over the wide temperature range. The methylcellulose content in CL:6MC is the most suitable for constructing a thermally stable hydrogel electrolyte. The hydrogel electrolyte (CL:6MC) also exhibits reversible rheological properties during heating and cooling cycles between 25 and 80 °C (Figure S12). The differential scanning calorimetry (DSC) curve of CL:6MC was analyzed by comparing it with that of CL:0MC (Figure S13). The CL:6MC shows only the smoothed exothermic peaks below 100 °C, corresponding the dynamic changes of matrix structure effected by hydrophobic association. The CL:0MC exhibits a significant exothermic peak at 87.7 °C, suggesting that the cellulose undergoes hydrolysis in the absence of methylcellulose repelling hydrated zinc ions.

The mechanical properties of the hydrogels were further investigated. At room temperature, the tensile tests of CL:xMC demonstrate a linear increase for the stress vs. strain curve, indicating a good elasticity of the hydrogels (Figure 2g). The tensile stress increases from CL:1MC to CL:6MC and decreases at CL:8MC, which is consistent with the rheological tests. For CL:6MC, the maximum stress and strain before failure are 126 kPa and 152%, respectively, indicating its high stretchability. The compressive stress-strain curves further demonstrate the excellent elasticity of the hydrogel and have the similar trend with the tensile curves (Figure S14). In the compression cycle tests, CL:6MC is able to recover well within 50% deformation (Figure S15). For

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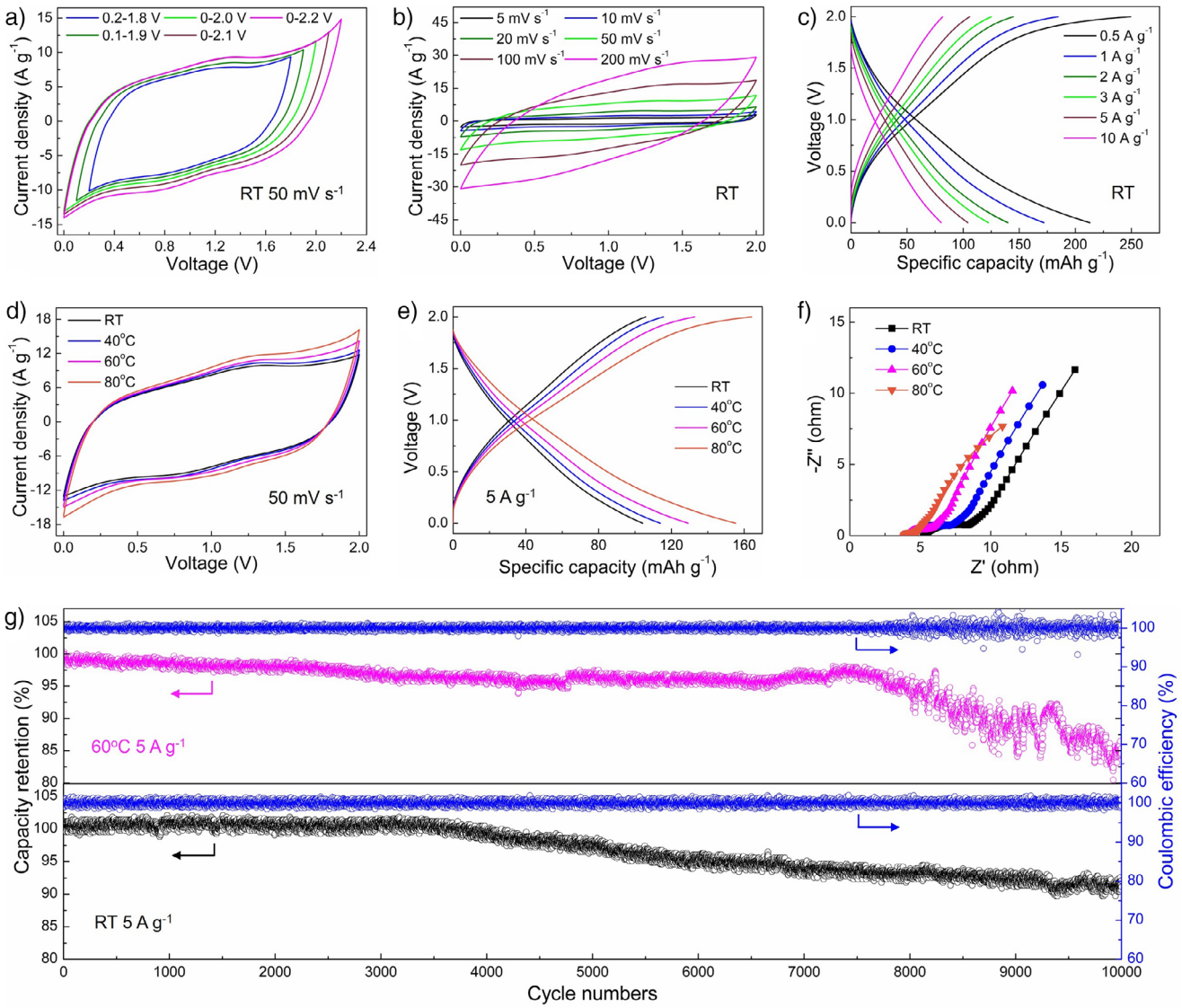

Figure 3. Electrochemical performance of the ZHS assembled from cellulose hydrogel electrolyte CL:6MC: a) CV curves with a scan rate of 50 mV s^{-1} under different voltage windows, b) CV curves under different scan rates, and c) GCD curves under different current densities at room temperature (RT); d) CV curves with a scan rate of 50 mV s^{-1} at different temperatures; e) GCD curves with a current density of 5 A g^{-1} at different temperatures; f) Nyquist plots at different temperatures; g) cycle testing with a current density of 5 A g^{-1} at room temperature and 60 °C.

higher temperatures at 40, 60 and 80 °C, the hydrogel can clearly remain its elasticity (G′ is significantly higher than G″ based on the rheological tests). Even at 80 °C, the CL:6MC can still be stretched to 50% strain with a tensile stress of 39 kPa (Figure 2h). The tensile elastic modulus was further calculated using the first 10% strain range in the tensile stress-strain curves.[49] CL:6MC has an elastic modulus of 130 kPa at room temperature, 38 kPa at 40 °C, 53 kPa at 60 °C, and 74 kPa at 80 °C. The elastic modulus of CL:6MC decreases from room temperature to 40 °C and increases from 40 to 80 °C, which indicates a non-monotonic change of the hydrogel matrix with temperature. As shown in Figure 2i, the compressive stress-strain curves of CL:6MC also show the similar trend. Such phenomenon can be explained by the temperature-dependent hydrophobic association effect. At room temperature, the hydrogel matrix is sustained mainly by coordinated cross-linking between cellulose and ions, and this cross-linking is broken with increasing temperature and exciting zinc ions.[31] As the temperature increases above 40 °C, the hydrophobic binding of methylcellulose intensifies, reconstituting the hydrogel matrix.[36] It is worth mentioning that CL:6MC also withstood freezing and remained flexible thanks to the suppression of hydrogen bonding between water molecules by highly concentrated ions (Figure S16).[41] The CL:6MC shows an ice point of −87.9 °C according to the DSC curve (Figure S17).

For energy storage application, we further investigated the ionic conductivity of the hydrogel electrolytes using EIS tests (Figure S18). As the methylcellulose content increases, the ionic conductivity of the hydrogel electrolytes decreases from CL:1MC to CL:8MC. For CL:6MC, the ionic conductivities at 25 (room temperature), 40, 60, 80, 100, and 120 °C are

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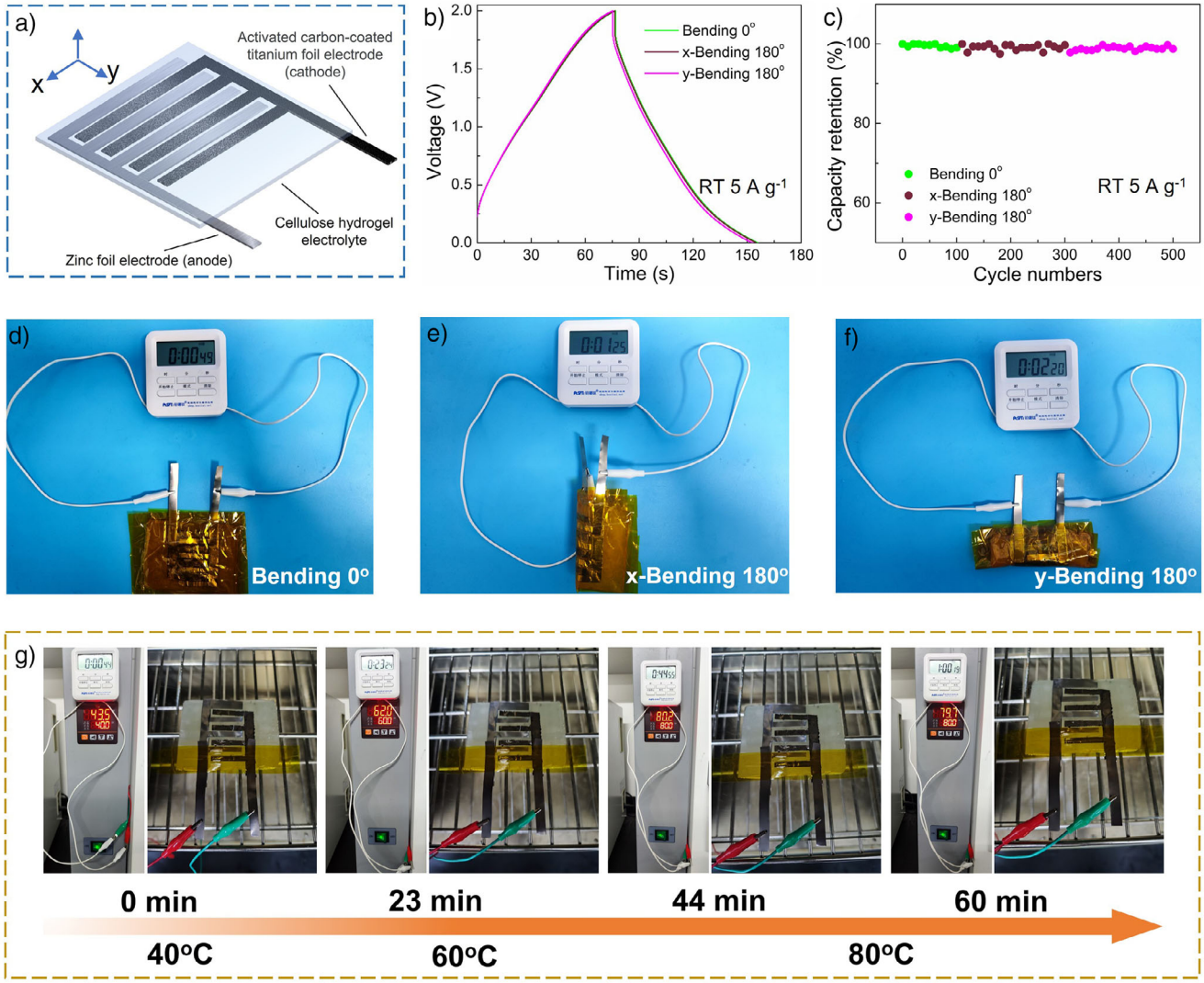

Figure 4. a) Schematic diagram of an in-plane ZHS with marking of the x- and y-directions, b) GCD curves of the in-plane ZHS with CL:6MC at different bending conditions (bending at 0°, bending at 180° in the x-direction, and bending at 180° in the y-direction), c) cycle testing at different static bending conditions, d) in-plane ZHS with CL:6MC working at 0° bending, e) 180° bending in the x-direction, and f) 180° bending in the x-direction, g) exposed in-plane ZHS with CL:6MC working at different high temperatures.

17.7, 16.8, 17.1, 18.6, 22.1, and 26.3 mS cm^{-1}, respectively. The slightly reduced ionic conductivity at a higher temperature of 40 °C compared to 25 °C can be attributed to the enhanced hydrophobic effect of methylcellulose. The ionic conductivity of the cellulose hydrogel electrolyte (CL:6MC) is competitive compared with other zinc-based hydrogel electrolytes (Table S1).[9,42,50,51] Using CL:6MC as a separator between the carbon-based cathode and the zinc foil anode, a ZHS was assembled. A stable electrochemical window from 0 to 2.0 V (versus Zn^{2+}/Zn) can be determined from the linear sweep voltammetry (LSV) curves (Figure S19). As shown in Figure 3a, the cyclic voltammetry (CV) curve of the ZHS shows a wide voltage window beyond 2.0 V without an obvious current surge. The electrochemical window of 0–2.0 V was used to further explore the electrochemical performance. The CV curve retains symmetrical rectangular shape as the scan rate increases from 5 to 200 mV s^{-1}, indicating the efficient supply and transport of charges by the cellulose hydrogel electrolyte (Figure 3b). The galvanostatic charge-discharge (GCD) tests were conducted with different current densities (Figure 3c). The ZHS has a high specific capacity of 212.9 mAh g^{-1} at 0.5 A g^{-1} and remains 80.3 mAh g^{-1} at 10 A g^{-1}. Combining the high specific capacity and wide voltage window, the ZHS with CL:6MC achieves a high energy density of 210.6 Wh kg^{-1} at a power density of 494.5 W kg^{-1}, and retains 68.7 Wh kg^{-1} at 8560.0 W kg^{-1} (the detailed Ragone plot can be found in Figure S20). The energy/power density of this device is comparable to that of other ZHSs with a hydrogel electrolyte.[52–56]

The electrochemical performance of the ZHS with CL:6MC was further conducted at different temperatures. The current density of the CV curve increases with temperature, indicating an increased capacity (Figure 3d and Figure S21). According to the GCD curves at 5 A g^{-1}, the ZHS

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with CL:6MC has the specific capacities of 104.2, 113.9, 129.2, 155.3 mAh g^{-1} at 25, 40, 60, and 80 °C, respectively, which corresponds to the coulombic efficiency of 98.5%, 98.6%, 97.3%, and 94.5% (Figure 3e). As the temperature increases from 100 to 120 °C, the specific capacity of the ZHS increases from 192.4 to 250.3 mAh g^{-1}, while the coulombic efficiency drops dramatically from 89.2% to 76.0% (Figure S22). The extremely high temperatures increase the likelihood of chloride ions undergoing electrolysis. As mentioned above, the ionic conductivity of the cellulose hydrogel electrolyte remains relatively unchanged over the temperature range from 25 to 80 °C, ensuring stable ion transport. The capacitive performance is affected mainly by the charge transfer process at different temperatures. As shown in Figure 3f, when increasing the temperature, the mid-frequency region of the Nyquist plot presents decreased charge transfer resistance, which corresponds to increased capacity (the equivalent circuit fitting of Nyquist plots can be found in Figure S23).[21] As the temperature increases to 80 °C, the Nyquist plot in the low-frequency part is bent toward the real axis, meaning that the diffusion process is significantly enhanced. The Nyquist plots at 100 and 120 °C show even lower charge transfer impedance and diffusion impedance (Figure S24). Considering that the practical application requires the device to function for a long time, the charging/discharging cycle test was performed at room temperature and 60 °C for comparison (Figure 3g). At room temperature, the ZHS with CL:6MC has a capacity retention of 91.1% after 10,000 cycles at 5 A g^{-1} and the coulombic efficiency always remains around 100%. The capacity decay may be related to the accumulation of by-products on the cathode electrode.[57,58] At the high temperature of 60 °C, the capacity retention after 10,000 cycles is still 85.4%. No observable appearance change was found for the CL:6MC hydrogel electrolyte at 60 °C for more than one week. The cycling in an environment with alternating temperatures between 25 and 60 °C demonstrates the stability of ZHS in the presence of thermal fluctuations (Figure S25). The thermal stability of the hydrogel electrolyte was key to the stability of the ZHS device at high temperatures. For temperatures higher than 80 °C, the effect of highly concentrated $ZnCl_2$ as a Lewis acid on cellulose increases dramatically.[46] The capacity retention of ZHS with CL:6MC after 1000 cycles at 80 °C is only 30.8% (Figure S26). The separation of electrode and hydrogel was observable. Further tests on the self-discharge properties were performed at room temperature and 60 °C after the ZHSs were fully charged to 2.0 V (Figure S27). After 120 h of settling, the voltage of the ZHS remains at 1.27 V at room temperature and 1.22 V at 60 °C. Compared with other studies focusing on high-temperature electrochemical devices, our study has produced excellent results (Table S2).[15,21,22,26,59,60] These include a wider electrochemical window, a higher specific capacity, better high-temperature stability (60 °C), and sustainability.

To validate the general feasibility of the hydrogel electrolyte, an in-plane ZHS was fabricated and subjected to testing under various mechanical and thermal conditions. As depicted in Figure 4a, the device is constructed with a carbon-based cathode and a zinc foil anode, both of which are seamlessly integrated into the cellulose hydrogel electrolyte (CL:6MC). The intrinsic adhesion properties of CL:6MC facilitate robust interfacial contact, ensuring mechanical integrity with an interdigital electrode configuration. The GCD curve at a current density of 5 A g^{-1} exhibits consistent morphology across different bending conditions (Figure 4b), indicating stable ion transport dynamics and interfacial stability. Furthermore, extended cycling under different static bending conditions demonstrates the long-term deformation resistance of the in-plane ZHS, as evidenced by its stable electrochemical performance over time (Figure 4c). The practical applicability of the in-plane ZHS under complicated condition was further verified by its ability to power a 1.5 V electronic meter under various bending conditions (Figure 4d–f). Additionally, the thermal stability of device was assessed by subjecting it to elevated temperatures of approximately 40, 60, and 80 °C (Figure 4g). Under these high-temperature conditions, the in-plane ZHS maintains stable, underscoring the high-temperature tolerance of the cellulose hydrogel electrolyte and the overall structural integrity of the energy storage system.

Conclusion

In summary, for the first time, we have proposed and realized a strategy by combining high salt concentration and hydrophobic associating matrix to obtain the hydrogel electrolytes suitable for high temperatures. An amphiphilic methylcellulose with temperature-dependent hydrophobic association can be stable in the highly concentrated $ZnCl_2$ (more than 65 wt%) with the help of cellulose. The cellulose hydrogel electrolyte is mechanically and electrically stable over a wide temperature range due to the synergy of cellulose and methylcellulose. As a pure bio-based hydrogel electrolyte, the hydrogel can be stretched to 152% at room temperature and 50% at 80 °C. Assembled with the cellulose hydrogel electrolyte, the ZHS has a wide voltage window of 2.0 V and achieve a high specific capacitor of 212.9 mAh g^{-1} at 0.5 A g^{-1}. Specially, the ZHS can exhibit enhanced high-temperature electrochemical performance. The ZHS has a stable 10,000 cycle life at 60 °C with an 85.4% capacity retention. This work opens up a general concept for the design of green hydrogel electrolytes with wide temperature range.

Acknowledgements

The authors thank the Natural Science Research Project of Guizhou Provincial Department of Education ([2022]041), Guizhou Provincial Basic Research Program (QKHJC-[2024]youth090), and Guizhou University Basic Research Project ([2023]03).

Conflict of Interests

The authors declare no conflict of interest.

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Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Cellulose • High temperature • Hydrogel electrolyte • Hydrophobic association • Zinc ion

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Manuscript received: September 24, 2025
Revised manuscript received: November 29, 2025
Manuscript accepted: December 10, 2025
Version of record online: ■■, ■■

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